Atharva Joshi

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Education

B.S. Mechanical Engineering, Minor: Mechatronics **GPA: 3.62/4.00**
The Pennsylvania State University, University Park, PA **Anticipated Graduation Date: December 2026**
Relevant Coursework: System Dynamics, Computer Architecture & Organization, Digital Logic, FEA & CFD, Robotics, Circuits

Professional Experience

Mechanical Engineering Intern, TE Connectivity 05/26 – 08/26
Middletown, PA

- Led an OSFP 224G test module from concept through preliminary release, creating Creo assemblies and GD&T drawings and completing PLM release workflows to unblock manufacturing.
- Performed statistical tolerance stack-ups for 3 OSFP 224G interference/clearance cases, validating with hand calculations to quantify failure rates (e.g., 306 PPM) to support DFM/NPI decisions.
- Designed a QSFP 224G connector organizer in Creo by adapting legacy parts, and verified a microscope-scale detent mechanism for a normal-force fixture to enable full fixture assembly.
- Authored an automation RFQ for a manual box unfolding/loading process, translating factory observations into mechanical, sensing, and HMI requirements for vendor quotation.

Mechanical Integrity Engineering Intern, Amphenol Communication Solutions 05/25 – 08/25
Valley Green, PA

- Designed a custom shock and vibration PCB test fixture in SolidWorks to avoid trace interference, and performed root-cause analysis via torque testing, identifying screw length variations as the primary failure mode.
- Executed electromechanical qualification tests, extracting low-level contact resistance (LLCR) data to generate statistical plots in Minitab and Excel, automating qualification reports via Overleaf.
- Developed and deployed a custom PowerApp to automate client failure requests, seamlessly linking data to SharePoint to streamline reliability testing intake workflows.

Lab Technician, OriginLabs, Penn State 08/24 – Present
University Park, PA

- Integrated electromechanical components (actuators, sensors, and controllers) into cohesive assemblies, ensuring seamless functionality between hardware mechanisms and software interfaces.
- Re-engineered a medical tablet holder using Design for Manufacturing (DFM) principles, reducing prototyping iterations by 40% through systematic design reviews and validation.

Undergraduate Research Assistant, Human-Centered Robotics Lab & S.H.A.P.E. Lab, Penn State 08/24 – Present
State College, PA

- Led the electromechanical integration of a SpotMicro quadruped robot, wiring actuators, sensors and a Teensy microcontroller; developed a C++ PID control algorithm for dynamic auto-balancing on up to 60-degree inclines.
- Conducted finite element convergence studies in Ansys and Abaqus on multi-material LPBF additive parts, optimizing mesh sizing to balance computational efficiency with stress and deformation accuracy.

Leadership Experience

Officer & Project Lead (Assistive Tech), American Society of Mechanical Engineers (ASME) 01/25 – Present
University Park, PA

- Directed a 15-member cross-functional team to design a visual navigation aid from scratch, integrating ultrasonic sensors and vibration motors within a custom SolidWorks-designed housing.
- Programmed Arduino Nano C++ logic utilizing a multi-sensor FOV for proximity and size detection, implementing personalized haptic feedback modes for visually impaired users.

Residence Life Student Leader, Penn State Residence Life 08/25 – 04/26
University Park, PA

- Mentored and supported a residential community of 500+ students across four halls, fostering connections through 6+ hours/month of engagement programs and individualized peer support.
- Partnered with 10+ campus offices and student organizations to design and deliver community-building initiatives, strengthening collaboration, communication, and problem-solving skills.

Projects

Portfolio: `https://wargodakj.github.io/`

- **SpotMicro Quadruped Robot**: Led the electromechanical integration of a quadruped robot, wiring actuators, sensors and a Teensy microcontroller; developed a C++ PID control algorithm for dynamic auto-balancing on up to 60-degree inclines
- **5-DOF Robotic Arm Kinematics**: Formulated and implemented analytical inverse kinematics algorithms in MATLAB to program a Braccio Robot for precise pick-and-place operations.
- **Five-Stage Pipelined MIPS CPU with Hazard Forwarding**: Designed and verified a five-stage pipelined MIPS-style CPU in Verilog with forwarding, load-use hazard stalling, delayed branch handling, Vivado synthesis/implementation, and successful FPGA bitstream generation on a ZyboBoard XC7Z010.
- **Tone Control/Karaoke Mixer Circuit**: Designed and simulated a five-stage audio processing system, featuring mixer/karaoke switching, tone control, and LED display.
- **Thermal Burner Analysis**: Developed a 1D finite-difference model in MATLAB utilizing a Gauss-Seidel SOR solver and bisection algorithm to evaluate thermal limits and maximum combustion gas temperatures of a Hencken burner's quartz casing.
- **Structural Vibration Analysis**: Analyzed harmonic, impulse, and forced responses of a 3-story shear building equipped with a pendulum tuned mass damper (PTMD) to evaluate dynamic structural damping performance.
- **Casting Process & Material Analysis**: Conducted a comparative study of conventional vs. 3D printed mold casting, analyzing thermal properties, defect rates, and material integrity for industrial applications.
- **Wheelchair-Mounted Dog-Food Dispenser**: Designed and prototyped a purely mechanical kibble dispenser for low-dexterity users. Utilized functional decomposition and selection matrices to develop a pull-slide mechanism across three iterative prototyping phases.
- **Manufacturing Robot Path Planning**: Modeled differential drive robot kinematics in Simulink and implemented Kalman Filters for state estimation and trajectory optimization.

Technical Skills

- **CAD & Simulation**: SolidWorks, PTC Creo, Windchill (PDM), Ansys (Structural, Thermal, Dynamic), Abaqus, Simulink.
- **Testing & Manufacturing**: GD&T, DFM/DFA, Rapid Prototyping, Vibration/Shock Testing, Root Cause Analysis.
- **Programming & Control**: C++, MATLAB, Minitab, Arduino DAQ.
- **Office Tools**: Microsoft Office Suite, SharePoint, Power Apps